

FAQs-2

Q. What is TF-IDF?

Ans.

TF-IDF:

- TF-IDF is a scheme to weight **individual** tokens.
- This weight is a statistical measure used to evaluate how important **a word is to a document in a collection of documents or corpus**.
- TF-IDF weighs words by the frequency of appearance within a document but penalizes words that appear across many documents.
- The concept is that words appearing many times in a single document are likely noteworthy, but they could also just be common words that appear frequently in all documents.

Read about **TF-IDF**: <https://www.geeksforgeeks.org/sklearn-feature-extraction-with-tf-idf/>

Q. What is Bag of Words (BoW) Model?

Ans.

Bag of Words (BoW):

- The Bag of Words (BoW) model is the simplest form of text representation in numbers.
- Like the term itself, we can represent a sentence as a bag of words vector (a string of numbers).
- The BoW model retains **neither information** on the **grammar of the sentences nor** on the **ordering of the words** in the text.

Read about **BoW**: <https://www.geeksforgeeks.org/bag-of-words-bow-model-in-nlp/>

Q. Describe the comparison of TF-IDF and Bag of Words.

Ans.

- **Bag of Words** just creates a set of vectors containing the count of word occurrences in the

document (reviews), while **the TF-IDF model** contains information on the more important words and the less important ones as well.

- **Bag of Words** vectors are easy to interpret. However, **TF-IDF** usually performs better in machine learning models.

Q. What are the drawbacks of using the Bag of Words (BoW) Model?

Ans.

- If the new sentences contain new words, then our vocabulary size would increase, and thereby, the length of the vectors would increase too.
- Additionally, the vectors would also contain many zeros, thereby resulting in a sparse matrix (which is what we would like to avoid)
- We are retaining no information on the grammar of the sentences nor on the ordering of the words in the text.

Q. How to remove numbers from string in the Pandas DataFrame column?

Ans.

- You can use the below code and define suitable function for it and iterate that over dataframe:

```
text = re.sub(r'\d+', '', text)
```

Q. How to install/download the Spacy model for English language?

Ans.

- You can use the below code inside the notebook:

```
!pip install https://github.com/explosion/spacy-models/releases/download/en\_core\_web\_sm-2.2.0/en\_core\_web\_sm-2.2.0.tar.gz
```